

Mohammad Refat

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RESEARCH INTERESTS

Keywords: Computational physics, Scientific computing, Statistical inference, Bayesian inference, Inverse problems, Time-series analysis, Data-driven modeling

Executive Summary: My research focuses on developing computational methods for extracting physical insight from complex data. Across projects in astrophysics, spectroscopy, Galactic archaeology, and time-domain astronomy, I have applied scientific computing, statistical inference, numerical modeling, and inverse methods to study complex physical systems. I am broadly interested in computational physics and in applying these approaches to challenging problems in astrophysics, computational biophysics, and soft condensed matter physics.

TEACHING

Adjunct Lecturer <i>Baruch College</i>	Aug 2025 – May 2026
– Introductory Physics Lecture (PHYS 2002)	Fall 2025, Spring 2026
– Introductory Physics Laboratory (PHYS 2003)	Fall 2025
– Physics II Laboratory (PHYS 3001)	Spring 2026

EDUCATION

City University of New York (CUNY) Graduate Center M.S. in Astrophysics Thesis: <i>Starspot Inference Using Light Curve Inversion Techniques</i>	2023–2025
CUNY Baccalaureate for Unique and Interdisciplinary Studies B.S. in Computational Astrophysics, <i>cum laude</i>	2019–2022
Bernard M. Baruch College	2017–2019

RESEARCH EXPERIENCE

Starspot Inference with Light Curve Inversion Techniques <i>American Museum of Natural History</i> <i>Advisor: Dr. Lucy Lu</i> – Investigated the feasibility and limitations of reconstructing stellar surface features from rotational photometric variability. – Applied light-curve inversion, time-series analysis, and Bayesian inference to study non-unique starspot configurations and model degeneracies. – Developed Python-based computational workflows for simulation, model fitting, and evaluation as part of an M.S. thesis on starspot inference.	Aug 2023 – Sep 2025
Towards Mapping Brown Dwarf and Giant Exoplanet Atmospheres	May 2021 – Aug 2023

American Museum of Natural History

Advisor: Dr. Johanna Vos

- Investigated atmospheric variability in brown dwarfs and giant exoplanets using rotational photometric observations.
- Applied time-series analysis and statistical modeling to characterize cloud-driven variability and infer atmospheric structure.
- Developed Python workflows for processing observational datasets and analyzing photometric light curves.

Chemodynamically Characterizing the Jhelum Stellar Stream with APOGEE-2

May 2020 – May 2021

Sloan Digital Sky Survey Faculty and Student Team

Advisor: Dr. Allyson Sheffield

- Investigated the chemical and dynamical properties of candidate stellar members of the Jhelum stellar stream using APOGEE-2 spectroscopy.
- Applied statistical analysis to identify stream members and characterize their chemical abundance patterns.
- Contributed to a refereed publication in *The Astrophysical Journal*.

Looking at Star Formation Through Chemistry

Aug 2018 – May 2020

American Museum of Natural History

Advisor: Dr. Allyson Sheffield

- Investigated stellar populations through spectroscopic measurements of chemical abundances.
- Processed and normalized echelle spectra and measured equivalent widths of spectral lines.
- Applied quantitative spectroscopic analysis techniques to characterize stellar chemical compositions.

Two-Point Statistics in the Star Forming ISM

Jun 2018 – Aug 2018

Flatiron Institute, Center for Computational Astrophysics

Advisor: Dr. Chang-Goo Kim

- Investigated statistical relationships between metallicity, gas dynamics, and star formation in the interstellar medium.
- Applied two-point statistical methods to analyze correlations within numerical simulation data.

PUBLICATIONS

Sheffield, Allyson A., Aidan Z. Subrahimovic, **Mohammad Refat**, et al. (May 2021). “Chemodynamically Characterizing the Jhelum Stellar Stream with APOGEE-2”. In: *The Astrophysical Journal* 913.1, p. 39. ISSN: 1538-4357. DOI: [10.3847/1538-4357/abee93](https://doi.org/10.3847/1538-4357/abee93). URL: <http://dx.doi.org/10.3847/1538-4357/abee93>.

Refat, Mohammad (2025). “Starspot Inference Using Light Curve Inversion Techniques”. Master’s thesis. CUNY Graduate Center. URL: https://academicworks.cuny.edu/gc_etds/6480/.

ACADEMIC PRESENTATIONS

Talks

- “Starspot Inference with Light Curve Inversion Techniques” 2025
CUNY Masters Graduation 2025 — Flatiron Institute, Center for Computational Astrophysics
- “Towards Mapping Brown Dwarf and Giant Exoplanet Atmospheres” 2022
240th American Astronomical Society Meeting — American Astronomical Society
- “Towards Mapping Brown Dwarf and Giant Exoplanet Atmospheres” 2021
Center for Computational Astrophysics / City University of New York / American Museum of Natural History Symposium — Flatiron Institute, Center for Computational Astrophysics
- “Chemodynamically Characterizing the Jhelum Stellar Stream with APOGEE-2” 2020
American Museum of Natural History REU Symposium — American Museum of Natural History
- “Where Were Stars Born in the Milky Way?” 2019
American Museum of Natural History REU Symposium — American Museum of Natural History
- “Two-Point Statistics in the Star Forming ISM” 2018
Center for Computational Astrophysics REU Symposium — Flatiron Institute, Center for Computational Astrophysics

Posters

- “Starspot Inference with Light Curve Inversion Techniques” 2025
245th American Astronomical Society Meeting — American Astronomical Society
- “Star Spots with Starry” 2024
22nd Cambridge Workshop on Cool Stars, Stellar Systems, and the Sun — University of California San Diego
- “Towards Mapping Brown Dwarf and Giant Exoplanet Atmospheres” 2023
Center for Computational Astrophysics Gotham Fest 2023 — Flatiron Institute, Center for Computational Astrophysics
- “Towards Mapping Brown Dwarf and Giant Exoplanet Atmospheres” 2023
241st American Astronomical Society Meeting — American Astronomical Society
- “Chemodynamically Characterizing the Jhelum Stellar Stream with APOGEE-2” 2021
237th American Astronomical Society Meeting — American Astronomical Society

TECHNICAL SKILLS

Programming	Python, MATLAB, Bash
Scientific Python	NumPy, SciPy, Pandas, Matplotlib, Astropy, Lightcurve, Astroquery
Computational Methods	Time-Series Analysis, Statistical Modeling, Inverse Problems, Bayesian Inference, Computational Modeling, Spectral Analysis
Software	Git, Linux, L ^A T _E X